

Opinion

Capabilities and challenges for the use of fluorescent RNAs in RNA dynamics research

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Visualizing RNA dynamics with high spatial and temporal resolution is key to understanding their biological function. Fluorescent RNAs (FRs), which are fluorescent protein (FP)-like entities comprising RNA aptamers and their cognate fluorogenic dyes, provide an attractive approach for visualizing RNAs in live cells. In this opinion, we present the current potential of emerging FRs for real-time and multiplexed fluorescence imaging of RNA dynamics at the single molecule level or with super-resolution and provide guidance for the methodological and experimental factors that need to be considered when performing RNA imaging by FRs, thus, increasing their effectiveness. We also provide perspectives and future directions for the development of FRs for advanced RNA imaging.

FRs revolutionize live cell RNA imaging

The expansion of genomics research has revealed that only a small fraction of RNA transcripts are translated into proteins, leading to intense efforts to elucidate the numerous ways in which **non-coding RNAs (ncRNAs)** (see [Glossary](#)) regulate cellular functions [1]. To better understand the full landscape of the influence of the spatiotemporal dynamics of ncRNAs on their biological function, approaches that enable real-time visualization of RNA dynamics in live cells are needed [2,3]. RNA-binding protein (RBP)-based genetically encodable approaches have significantly advanced our understanding of RNA dynamics through imaging [4–9]. However, the possible perturbation of RNA stability and trafficking by multiple protein binding remains to be resolved [10,11]. Therefore, a protein binding-independent approach may overcome the aforementioned limitations and facilitate new avenues for RNA imaging.

FRs, which are FP-like entities comprising **RNA aptamers** and their cognate **fluorogenic dyes**, provide an attractive approach for visualizing RNAs in live cells ([Figure 1A](#)). In FRs, the fluorogenic dye is dimly fluorescent by itself and becomes strongly fluorescent only upon binding to its cognate RNA aptamer; this enables RNA imaging with high signal-to-background (S/B) ratios [12,13]. To date, dozens of FRs have been developed and categorized into **molecular rotor (MR)**-based [12,14–29], **fluorophore-quencher conjugate (FQC)**-based [30–36] and **spirocyclic fluorophore (SF)**-based FRs [37–39] ([Figure 1](#)). Over the past 5 years, FRs with much improved brightness and stability have been reported, which have enabled robust imaging of diverse RNA species in live cells at the single molecule level or with super-resolution; this research has moved FRs from a concept to practical use ([Figure 1B](#)). Moreover, multiplexed fluorescence imaging of RNA using orthogonal FRs has also been accomplished, enabling the simultaneous imaging of multiple RNA species for the examination of the interactions or functional correlations among them ([Box 1](#)) [27–29,40]. In addition, FRs can be engineered into probes for the labeling of endogenous RNA without exogenous fusion with FR tags ([Box 2](#)) [25,41–45]. For a detailed summary of currently available

Highlights

Visualizing RNA dynamics with high spatial and temporal resolution is key to understanding their biological function.

Fluorescent RNA (FR) has emerged as a promising tool for real-time imaging of noncoding and coding RNA at the single molecule level or with super-resolution.

Orthogonal FRs can be used for multiplexed RNA imaging, allowing simultaneous visualization of multiple RNA species in live cells.

FR-derived probes can image endogenous RNA without genetic manipulation.

Advances in FRs provide valuable tools not only for various fields in basic life science research in which RNAs are involved, but also for facilitating the development of advanced technologies for biosensing and diagnosis.

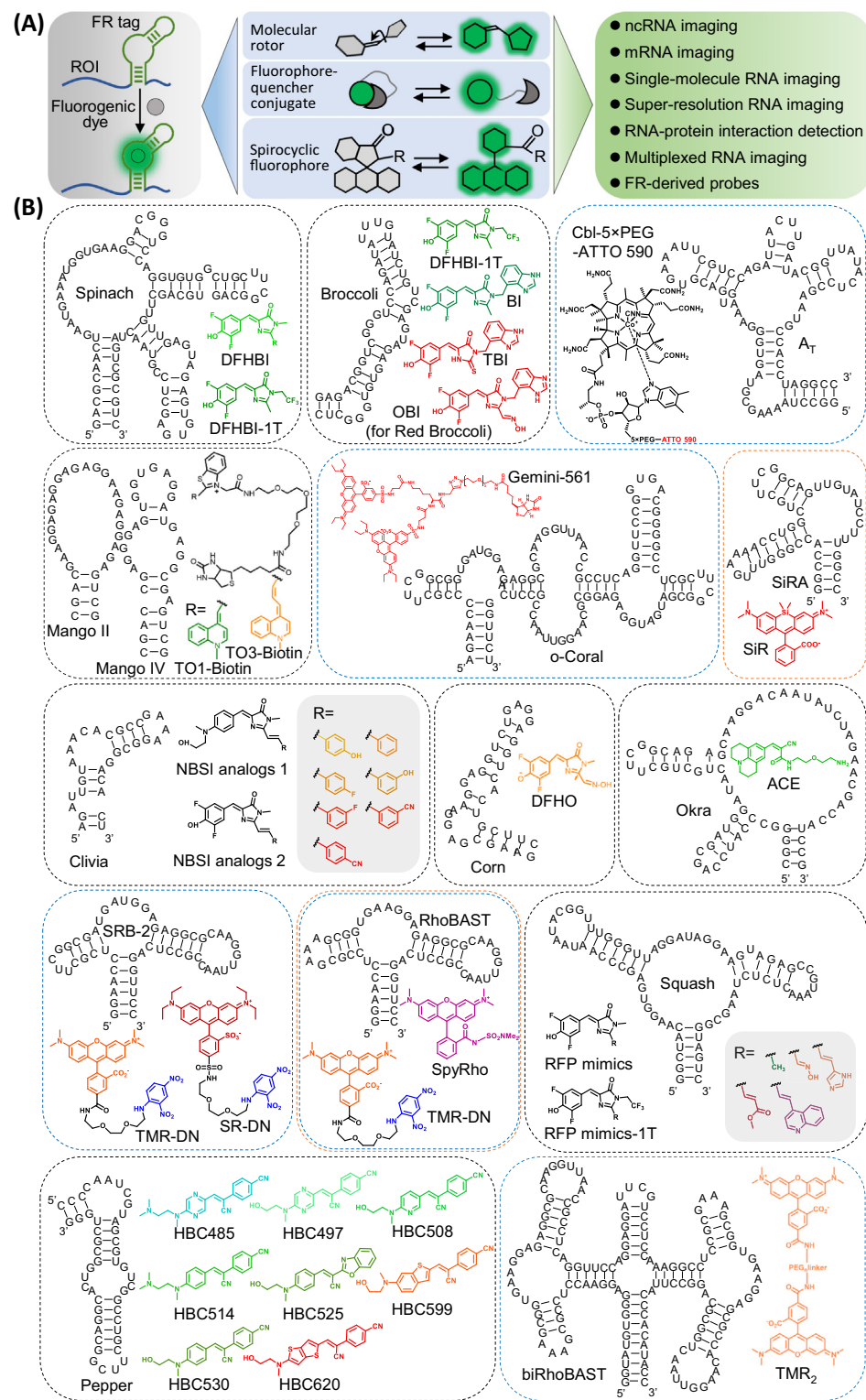
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Glossary

Bioluminescence resonance energy transfer (BRET):

electromagnetic phenomenon in which quantum energy is transferred nonradiatively from a luminescent donor to a fluorescent acceptor within close proximity.

Cajal body: spherical structure found in the nucleus of highly proliferative cells; it is closely associated with the nucleolus and contains proteins and RNAs involved in various nuclear processes.

Fluorescence resonance energy transfer (FRET):

electromagnetic phenomenon in which quantum energy is transferred nonradiatively from an excited donor fluorophore to an acceptor molecule within close proximity.

Fluorescent RNAs (FRs): RNA mimics of fluorescent proteins that are RNA aptamers developed *in vitro* and capable of binding fluorogenic dyes and turning on their fluorescence.

Fluorogenic dyes: fluorescent dyes, the fluorescence intensity of which changes in response to changes in the polarity, viscosity, or molecular order of their microenvironment.

Fluorogenicity: ability to transform the state of a fluorophore from a 'dark' to a 'bright' state, typically caused by a trigger event, such as target binding or a laser pulse.

Fluorophore-quencher conjugate

(FQC): complex comprising a fluorophore conjugated with a quencher, where the direct interaction between the fluorophore and quencher induces the formation of a nonfluorescent intramolecular complex.

miRNAs: a class of small ncRNAs that are typically 21–25 nucleotides in length and function in the post-transcriptional regulation of gene expression.

Molecular rotor (MR): type of fluorophore comprising an electron donor and an acceptor linked by π conjugation, which undergoes a twisting motion that leads to extremely low fluorescence quantum yields in the excited state.

Noncoding RNA (ncRNA): category of RNA that does not encode proteins but has important roles in diverse cellular processes.

RNA aptamer: short oligonucleotide sequence that is highly structured and can bind to targets with high affinity and specificity.

Box 1. Multiplexed imaging of RNA by orthogonal FRs

Multiplexed imaging to simultaneously visualize multiple RNAs is important for understanding the correlations among the different RNA species that are involved in cellular processes. FRs capable of multiplexed imaging need to satisfy the following two criteria: (i) they should be orthogonal to avoid cross-interaction among the aptamers and fluorogenic dyes; and (ii) they should have minimal spectral overlap to reduce false-positive signals caused by bleed-through (see Figure 3A,B in the main text). To date, combinations of SRB-2/DNB [31], Pepper/Okra [28], Pepper/Clivia [27], Pepper/Clivia/Myosotis [29], Squash/Broccoli [50], biRhoBAST/biSiRA [35], Pepper/RhoBAST [39], and Broccoli/Chilli/Corn/DNB/Pepper [40] have been reported to have good biorthogonality, and several have been successfully used for multiplexed RNA imaging. For example, Zuo *et al.* used Pepper and Okra for dual-color super-resolution imaging of *ACTB* and *GAPDH* mRNAs in SGs, revealing non-uniform and distinct distributions of the different RNAs throughout the SGs [28]. Given that several Clivias share excitation spectra similar to those of many Peppers but have large Stokes-shifted emissions, Clivia and Pepper enabled single-excitation two-emission dual-color imaging of U6 snRNA and *ACTB* mRNA in live mammalian cells. This single-wavelength excitation for multiplexed imaging significantly reduces the image acquisition time and phototoxicity, provides a better opportunity for the study of multiple processes more quickly or for longer duration and lowers the cost of instrument investment; these factors are particularly important for two-photon microscopy or axially swept light-sheet microscopy [79–82]. Recently, Zheng *et al.* developed a multiplexed sequential imaging approach named seqFRIES; this approach solved the current spectral overlap problem and was successfully used for semiquantitative four-color RNA imaging in live cells by using four orthogonal FRs [40]. With the development of more orthogonal FRs, we anticipate that five- or even six-color RNA imaging can be attained in the near future.

rRNAs: ncRNAs that self-associate with ribosomal proteins to form ribosomes and have critical roles in protein synthesis.

Small interfering RNAs (siRNAs): small antisense RNAs ~20–25 nucleotides in length that interfere with the expression of particular genes and mRNA translation.

Spirocyclic fluorophore (SF): type of fluorophore with a spirocyclic molecular structure that exploits the open–closed equilibrium for fluorescence activation.

Stress granule (SG): type of membraneless cytoplasmic RNA granule that forms from translationally stalled mRNAs in response to various stresses.

tRNAs: small ncRNAs that participate in protein synthesis by delivering amino acids to the ribosome.

U-body: a cytoplasmic RNA granule that is a site for small nuclear ribonucleoprotein assembly.

FRs, we refer readers to several recent reviews of the development of FRs and their photophysical properties [13,46,47].

Here, we discuss the capabilities and challenges of state-of-the-art FRs to examine the influence of RNA on cellular function using FR-based imaging. We summarize recent progress in applying FRs for real-time, single molecule, super-resolution and multiplexed fluorescence imaging of RNA and for the detection of RNA–protein interactions in live cells. Finally, we offer guidance for the methodological and experimental factors that need to be considered when performing RNA imaging by FRs, and provide perspectives and future directions for the development of FRs for advanced RNA imaging.

Uses of FRs in RNA dynamics research

Over the past decade, FRs have shown great potential in many aspects of RNA dynamics research. Real-time visualization of ncRNAs and mRNAs has been attained by simply fusing FR tags to the 5′, 3′, or internal regions of target RNAs. Several FRs with high brightness and photostability enable single molecule and super-resolution RNA imaging. In addition, RNA–protein interactions can be detected by FR-based approaches, allowing direct and sensitive monitoring of such interactions in live cells (Table 1). These applications strengthen the utility and versatility of FRs for advanced RNA imaging.

Small ncRNA imaging

Small ncRNAs, such as **tRNAs**, some **rRNAs**, and **small interfering RNAs (siRNAs)**, are involved in the regulation of numerous biological processes [1]. In cells, small ncRNAs usually interact with RBPs and other RNA molecules, which affects their ability to be bound by fluorescence *in situ* hybridization (FISH) probes or molecule beacons (MBs) [48]. Owing to their small sizes and high S/

Figure 1. Overview of different types of fluorescent RNA (FR) and their applications in RNA dynamics research.

(A) In molecular rotor (MR)-type FRs, the dyes undergo a twisting motion that leads to extremely low fluorescence. The binding of RNA aptamers can inhibit the twisting motion, leading to markedly increased fluorescence of the dyes. In fluorophore-quencher conjugate (FQC)-type FRs, the fluorescence of dyes is quenched through the contact between the fluorophore and quencher. The binding of the fluorophore or quencher to an aptamer can physically separate the two moieties, thus restoring the fluorescence of the fluorophore. In spirocyclic fluorophore (SF)-type FRs, the dyes exhibit an open–closed equilibrium between a noncolored spirolactone and a fluorescent zwitterion. The fluorescence increases upon binding of an aptamer to the dyes. (B) FRs used in live cell RNA dynamics research. Structure of the fluorogenic dyes and predicted secondary structures of the aptamers in FRs are shown. Black, blue and orange-dashed boxes represent MR-based, FQC-based, and SF-based FRs, respectively. Abbreviation: ncRNA, noncoding RNA.

Box 2. Imaging of endogenous RNA with FR-derived probes

For visualization of RNA using FRs, the protocol needs to include genetic manipulation to fuse an FR tag to the target RNA; this aspect is difficult for imaging endogenous RNA because of the need to modify genomic DNA. To overcome this limitation, FR-derived probes have been developed for specific RNA labeling using sequence hybridization. The design principle behind these probes is to use unstable FRs such that they are nonfluorescent and unable to bind their fluorogenic dyes. Only hybridization with the target RNA enables reassembly of FRs into their correct configurations capable of binding and activating their fluorogenic dyes. To date, several FRs, including SRB-2 [83], Broccoli [44], Pepper [40–42,45,84], RhoBAST [85], and Squash [25], have been engineered into probes for imaging ncRNAs and mRNAs in live cells (Figure 1). Compared with original FRs, FR-derived probes do not need to be fused to the target RNAs; this aspect would be particularly useful for imaging endogenous RNAs since it avoids the time-consuming and laborious steps to modify the genomic DNA. However, limited by the hybridization efficiency and nonspecific activation (hybridization with RNA molecules with high sequence similarity to the target RNA) of the probes, FR-derived probes typically have lower sensitivity and S/B ratios compared with the original FRs.

Three considerations need to be taken into account in the design of efficient FR-derived probes. The first is the accessibility of target RNA, which affects the hybridization efficiency of the probes. Typically, the regions in target RNA that do not form stable secondary structures and do not participate in protein or RNA binding can be selected as the hybridization sites. The second is the energetic calculation. Two distinctive phases of energy are needed for the hybridization of the probe to target RNA. The first phase involves opening the hybridization sites of the target RNA and probe, and the second phase involves hybridization between the target RNA and probe. In practice, the first phase of energy needs to be less than the second phase of energy [42,48]. The third consideration is nonspecific activation. Alignment of the hybridization sequence with the entire transcriptome of the host cell is recommended to reduce nonspecific activation by hybridization with RNAs with high sequence similarity to the target RNA.

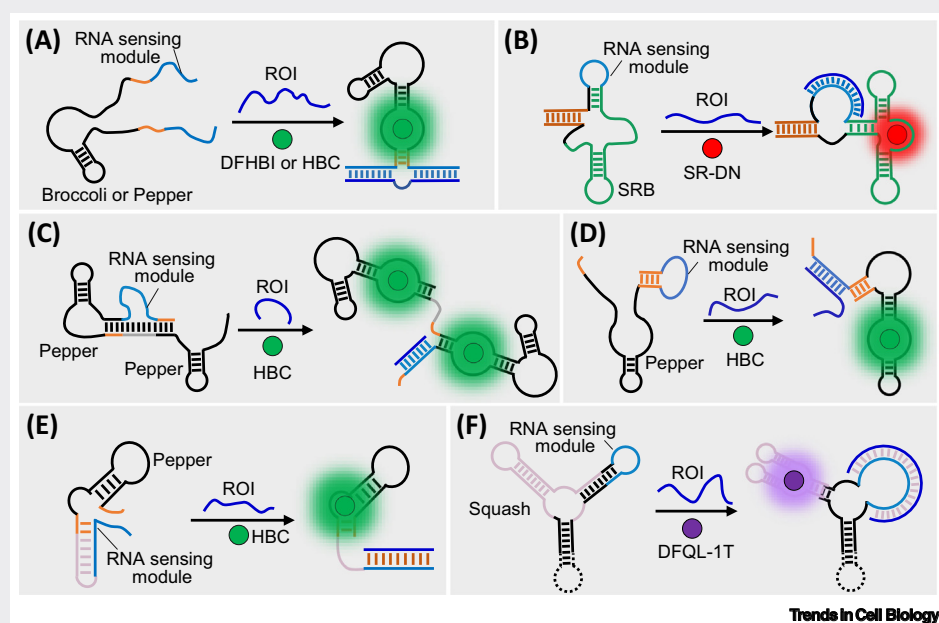


Figure 1. Imaging of endogenous RNA with fluorescent RNA (FR)-derived probes. (A,C–E) Pepper-, (A) Broccoli-, (B) SRB-, and (F) Squash-derived probes allow imaging of endogenous RNA without genetic manipulation of the RNAs of interest (ROIs). Hybridization of the probes occurs through complementary pairing of the RNA-sensing modules in probes with the ROIs.

B ratios, FRs have provided a gap-filling tool for visualizing the dynamics of small ncRNAs in live cells. The first attempt was made by Paige *et al.*, who used Spinach to tag 5S rRNA and monitored the dynamics of the rapid sucrose-induced increases in total 5S-Spinach levels in live mammalian cells [12]. Several FRs have also shown the capacity for live cell imaging of small ncRNAs, such as 5S rRNA [20,26,36,49,50], 7SK [18,26,27,35], U1 [27,33], U6 [26,27,35,49], and mgU2-47

Table 1. Uses of FRs in RNA dynamics research^a

FR	Small ncRNA imaging	mRNA imaging	Single molecule RNA imaging	Super-resolution RNA imaging	RNA-protein interaction	Endogenous RNA imaging	Refs
Spinach:DFHBI	5S	●	○	○	○	○	[12,86]
Spinach2:DFHBI	5S, 7SK	○	○	○	○	○	[18]
Pepper:HBCs	5S, 7SK, U6	●	○	SIM	●	●	[26,40–42,45,84]
Clivia:HBSIs	U6, U1, 7SK, mgU2-47	○	●	SIM	●	○	[27]
Okra:ACE	○	●	●	SIM	●	○	[28]
biRhoBAST:TMR ₂	U6, 7SK	●	●	○	○	○	[35]
RhoBAST:SpyRho	○	●	○	SMLM, STED	●	○	[39]
RhoBAST:TMR-DN	○	●	○	SMLM	○	●	[34,85]
SRB-2: SR-DN/TMR-DN	○	●	○	○	○	●	[32,83]
Squash: DFHO/DFQL-1T	5S	○	○	○	○	●	[25,50]
Broccoli:BI	○	●	●	○	○	●	[23,44]
A ₇ :Cbl-5×PEG-ATTO 590	U1	●	○	○	○	○	[33,87]
Mango II:TO1-B	5S, mgU2-47	○	○	○	○	○	[17,49]
Mango IV:TO1-B	U6, 5S	●	●	○	○	○	[49]
SiRA: SiR	○	●	○	STED	○	○	[37]
o-Coral:Gemini-561	○	●	○	○	○	○	[36]
Myotosis:DBT	○	●	○	○	○	○	[29]

Abbreviations: SIM, structured illumination microscopy; SMLM, single-molecule localization microscopy STED, stimulated emission depletion.

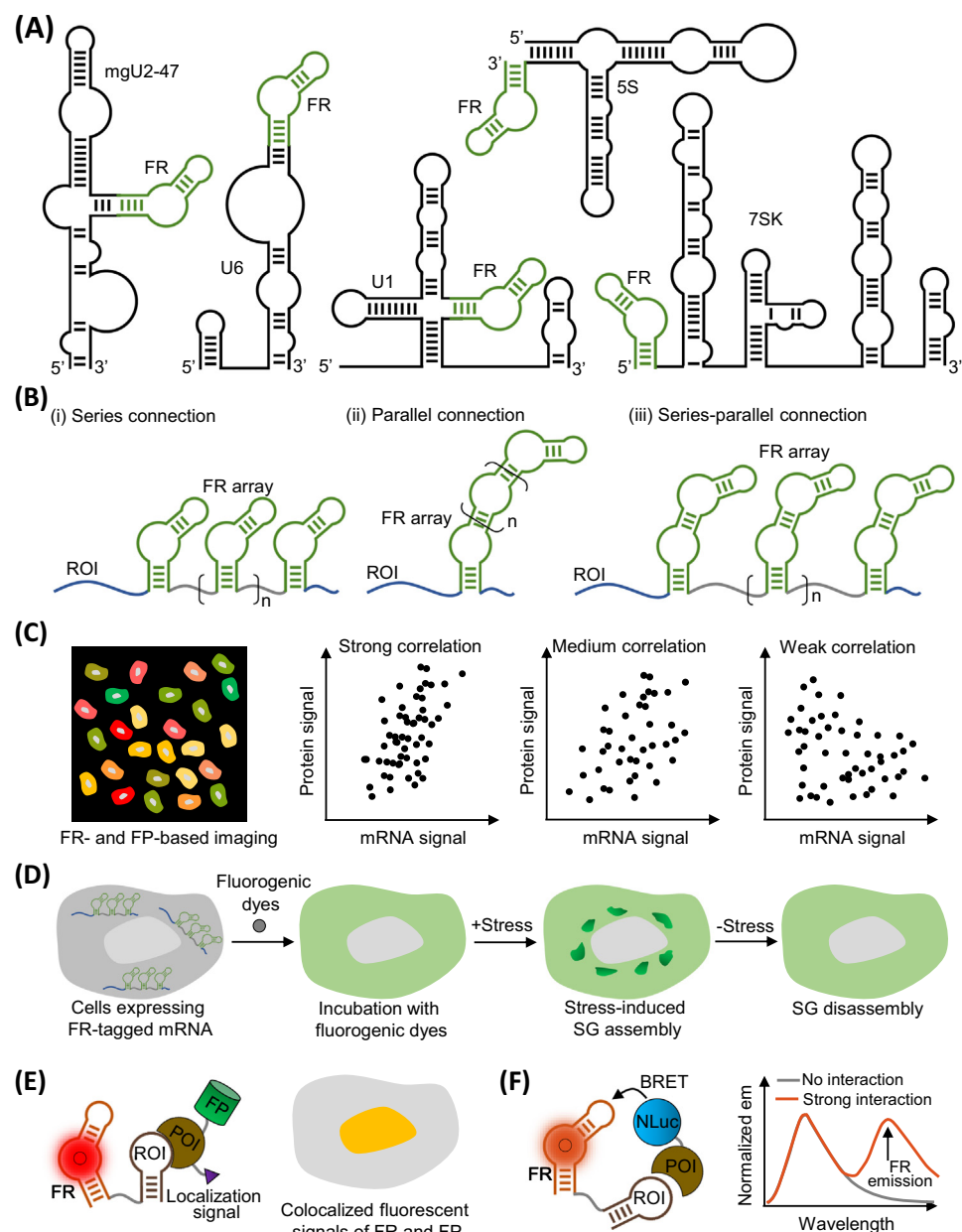
^a○, not reported; ●, reported.

[27,49] small nuclear (sn)RNAs (Figure 2A). Notably, Clivia, which is one of the smallest FRs, containing only 30 nucleotides, is particularly useful for tagging diverse small ncRNAs since its small volume can minimize potential interference with the localization and functionality of small ncRNAs. By using Clivia-tagged U1, Jiang *et al.* successfully monitored **U-body** dynamics after the cells were treated with thapsigargin. Interestingly, they reported that the induced U-bodies remained stable after thapsigargin withdrawal but disappeared or became much smaller after the cells entered mitosis [27]. In addition to rRNA and snRNA, small regulatory ncRNAs, including **miRNAs** and siRNAs, have also been visualized using FR-derived probes (Box 2). Compared with marker protein-based approaches [51], FRs provide a more straightforward way to directly reflect the true distribution and behaviors of small ncRNAs.

mRNA imaging

Compared with many small ncRNAs that are transcribed from RNA polymerase (Pol) III, mRNAs transcribed from RNA Pol II typically have a much lower abundance than small ncRNAs [26]; thus, visualizing mRNAs with adequate S/B ratios in mammalian cells is challenging. To overcome this issue, tandem repeats of FR tags are fused to a single RNA transcript to produce amplified fluorescence signals [20,26]. Three strategies are typically used to design tandem repeats of FRs: series connection [26,34,35], parallel connection [26], and series-parallel connection [23] (Figure 2B).

Similar to the repetitive arrays of MS2 harpins, the tandem repeats of FRs are usually fused to the 3'-untranslated regions (UTRs) of mRNAs, which have minimal interference with the



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Figure 2. Capabilities for fluorescent RNAs (FRs) in live imaging of RNA dynamics. (A) Schematic of FR-tagged different small noncoding (nc)RNAs for live-cell imaging. (B) Different strategies for the design of tandem repeats of FRs. In a series connection, the 5' end of one molecule FR tag is fused to the 3' end of another molecule FR. In a parallel connection, one FR molecule is fused to the terminal stem loop of another FR molecule. A series-parallel connection is a combination of the series and parallel connections. (C) Simultaneous labeling of the mRNA and protein levels to reveal strong or weak correlation between the protein and mRNA levels. (D) FR-based real-time monitoring of mRNA dynamics. FR-tagged mRNAs assemble to form large aggregates of stress granules (SGs) under stress conditions. The aggregates disappear after the withdrawal of stress. (E) Live cell imaging of RNA-protein interaction through detection of the colocalized fluorescent signals of FR and fluorescent protein (FP). (F) FR-based live cell imaging of RNA-protein interaction through bioluminescence resonance energy transfer (BRET).

translation and stability of mRNA molecules. To date, the tandem repeats of SRB-2 (16×) [32], Broccoli ($n = 24$ or 48), Pepper ($n = 4$ or 8) [26], o-Coral ($n = 1$) [36], riboswitch tag A ($n = 4$) [33], RhoBAST ($n = 2, 4, 8$, or 16) [34,39], biRhoBAST ($n = 2$ or 8) [35], and Okra ($n = 2, 4, 6$, or 8) [28] have been successfully used to visualize mRNA in live mammalian cells. Notably, both o-Coral and biRhoBAST were designed through avidity-based binding of dimeric aptamers to their fluorogenic bivalent dyes; thus, each contains two molecules of the monomeric FR through series connections [35,36]. These FR arrays enable the simultaneous detection of mRNA and protein levels in single cells through flow cytometry or an imaging method. For example, Chen *et al.* studied mRNA translation control using Pepper FR and an FP to simultaneously label mRNA and protein levels, respectively. They reported that different mRNAs had distinct affinities for the ribosome and heterogeneity in translation efficiency (Figure 2C). Interestingly, more marked heterogeneity was found in a few cancer cell lines, reflected by the extremely weak correlation between the protein and mRNA levels [36]; this research provided new perspectives for the diagnosis and treatment of cancers.

FRs also allow real-time monitoring of mRNA dynamics in live cells. Zuo *et al.* used Okra to monitor mRNA trafficking to **stress granules (SGs)**. After treatment with sodium arsenite (NaAsO_2), the Okra-tagged *ACTB* mRNA assembled to form large aggregates of SGs within 60 min; these aggregates became progressively smaller and eventually disappeared within 120 min after the withdrawal of NaAsO_2 (Figure 2D). The authors also reported that ATP has an important role in the assembly, disassembly, and even dynamics of SGs after assembly [28]. Compared with FISH or marker protein-based approaches [52], FRs provide a more straightforward way to study the dynamics of mRNA molecules in real time in live cells. In addition, FR-derived probes enable labeling and imaging of endogenous mRNA molecules without genetic modification of the genome (Box 2).

Single molecule RNA imaging

Over the past two decades, the MS2 system has been the gold standard for single molecule RNA imaging, enabling the examination of RNA behavior in live cells [2]. As described in the preceding text, the fluorescence signals of FRs can be amplified by fusing multiple FR tags to form FR arrays, enabling single molecule RNA imaging with improved sensitivity. The first attempt was carried out by fusing a 24× Broccoli or 48× Broccoli array to the 3'-UTR of β -actin mRNA. Numerous mobile fluorescent puncta were observed in the cytoplasm, which effectively colocalized with the fluorescent signals from the MS2-MCP system; these results indicated that the mobile puncta clearly represented single mRNA transcripts [23]. Since then, 24× Mango II, 8× Clivia, and 8× Okra repeats have also shown the ability to image RNAs at the single molecule level without affecting their inherent localization patterns [17,27,28]. Notably, the dimeric FR biRhoBAST, with impressively high binding affinity for its bivalent fluorogenic dye, has been shown to be particularly useful for single molecule tracking of RNA molecules in live cells with minimal interference in RNA localization, translation, and degradation [35]. When 8× biRhoBAST and 24× MS2 hairpin repeats were simultaneously fused to the 3'-UTR of mRNA, biRhoBAST presented a higher S/B ratio compared with the MS2 system, even with fewer fluorophores (16 TMR vs. 48 GFP) [35]. Together, the FR-based protein-free approach has shown great potential as an excellent tool for the study of RNA dynamics at single molecule resolution.

Super-resolution RNA imaging

Super-resolution imaging approaches, such as stimulated emission depletion (STED) microscopy, structured illumination microscopy (SIM), and single-molecule localization microscopy (SMLM), require fluorescent probes with high brightness and photostability [53]. To date, several FRs with superb brightness and photostability have been successfully applied in super-resolution RNA imaging. Pepper:HBC620, which has *in vitro* brightness and *in vivo* photostability far

exceeding that of mCherry, was the first to be used in SIM-based super-resolution imaging of RNA in live mammalian cells [26]. Thereafter, Clivia and Okra FRs were also used for SIM imaging of small ncRNAs and mRNAs, revealing non-uniform distributions of these RNAs in RNA granules, including **Cajal bodies**, U-bodies, and SGs [27,28]. These SIM images may be useful for studying the relationship between the structure and functionality of diverse cellular RNA granules. In addition to SIM, FR-based STED and SMLM imaging of RNA have also been reported [34,37,39]. In 2019, Wirth *et al.* first used SiRA FR for STED imaging of mRNA in live bacterial cells, achieving a significantly increased resolution of ~40 nm [37]. Another two FRs, RhoBAST:TMR-DN and RhoBAST:SpyRho, were subsequently applied in SMLM and STED imaging of the structural details of the nuclear CGG trinucleotide repeat-containing fragile-X mental retardation (*FMR1*) mRNA aggregates in live mammalian cells [34,39]; the results showed greatly enhanced spatial resolution with 25-nm localization precision [39]. Recently, photoactivable FRs were developed [54] that could be useful tools for super-resolution RNA imaging similar to the wide use of photoactivable FPs in diffraction-limited and super-resolution imaging [55]. Together, the subcellular and subnuclear structures of RNA can be readily visualized by FR-based super-resolution imaging in live cells with excellent localization precision; this imaging will be useful for the study of correlations between the localization and functionality of RNA.

RNA–protein interaction imaging

RNAs and proteins are interconnected biomolecules that can promote each other's life cycles and functions through physical interactions [56]. Although several approaches have been used to study the interactions between RNAs and proteins, they are limited by requiring lysis of the cells, small dynamic changes in **fluorescence resonance energy transfer (FRET)** itself, or the high background fluorescence of the dye [56–60]. These limitations can be overcome by FR-based approaches. In one approach, an FR tag is fused to the RNA of interest (ROI), and an FP tag is fused to the protein of interest (POI), which is localized to a specific subcellular structure. If the ROI interacts with the POI, well-colocalized fluorescent signals from the FR and FP can be clearly observed (Figure 2E). As a very preliminary proof-of-principle example, the well-studied MS2–MCP interaction was validated in live cells using different FRs [26,27,39]. However, this approach can reflect only whether the RNA and protein interact, and strong binding is difficult to differentiate from weak binding between the RNA and protein. This limitation was overcome by another FR-based approach. In this approach, the Clivia FR, which has a unique spectral property of large Stokes shift and can serve as an ideal **bioluminescence resonance energy transfer (BRET)** acceptor for NanoLuc (NLuc) luciferase, is fused to the 5', 3' or terminal stem-loop (SL) of ROI, and the POI is fused to the N or C terminus of NLuc. If the ROI and POI do not interact, only one signal emitted from the luciferase can be detected in the presence of the NLuc substrate furimazine. If the ROI and POI interact, BRET occurs between the luciferase and Clivia, and an additional signal emitted by Clivia can be detected (Figure 2F). Using this approach, several exogenous and endogenous RNA–protein interactions were successfully detected in live cells [27]. More importantly, the relative binding affinities among RNA, proteins, and their variants could be easily distinguished by measuring their BRET efficiencies [27].

Since BRET efficiency is closely related to the relative distance between the NLuc and Clivia dipoles, this approach can also be used to understand more physiologically relevant RNA–protein interactions, such as the interaction between U1 snRNA and U1-70K protein [27]. Unlike fluorescence-based approaches, this BRET-based approach does not require exogenous illumination from external light; thus, the background fluorescence caused by the cellular components is minimized, and cross-excitation of the acceptor is prevented [61]. Moreover, unlike traditional biochemical methods that are performed in solution [56], FR-based approaches provide straightforward methods for the direct detection of RNA–protein interactions in cellular environments.

Brief guide for FR-based RNA imaging

As discussed in the preceding text, FRs have provided numerous opportunities for visualizing RNA dynamics in live cells. Although they are simple and promising, several methodological and experimental factors need to be considered when performing FR-based RNA imaging. Here, we provide a brief guide in terms of the considerations of S/B ratio, cellular brightness, and background, with the aim of enhancing the effectiveness of FRs.

Cellular brightness

The cellular brightness of an FR tag is mainly determined by four factors: (i) *in vitro* brightness (extinction coefficient and quantum yield); (ii) nuclease resistance of FR tag; (iii) folding efficiency of FR tag in the cellular environment; and (iv) effective concentration of the fluorogenic dye (Figure 3C). Typically, the *in vitro* brightness is measured under optimal conditions (e.g., nuclease free, and appropriate ion concentrations and temperature). However, the cellular environment differs significantly from the *in vitro* scenario, which sometimes makes FRs less bright in cells compared with *in vitro*. For example, Broccoli:BI has a comparable *in vitro* brightness to that of Okra:ACE but has a <30-fold lower cellular brightness [28]. Moreover, it is usually difficult for high-molecular-weight or charged dyes to cross the plasma membrane, making the effective concentration of the dyes inside cells lower than required, which leads to lower cellular brightness. For example, the fluorogenic dye in the Riboglow platform comprises a cobalamin quencher and a fluorophore linked by polyethylene glycol (PEG) units. The high molecular weight of the compound and PEG linker reduce the permeability of the dye, making it necessary to use a bead-loading method to increase its effective concentration (Figure 1B) [33]. Among the currently available FRs, Pepper:HBC485 (cyan) [26], Okra:ACE (green) [28], RhoBAST:SpyRho (orange) [39], Pepper:HBC620 [26], and Squash:DFQL-1T [25] outperform other spectrally matched FRs in terms of cellular brightness. However, brightness is not the only consideration when selecting a suitable FR for RNA tagging. For example, avidity-based FRs comprising dimeric aptamers and their bivalent fluorogenic dyes have remarkably high cellular brightness, but the markedly increased size of the RNA tag may limit their use for imaging small ncRNAs, such as U1 [35,36].

Background fluorescence

The background fluorescence of FR-based imaging is caused mainly by the dye itself and its non-specific interaction with cellular components. Molecule rotor dyes typically have high **fluorogenicity** that can be >10 000-fold fluorescence turn-on ratios; thus, their background fluorescence in unbound states is lower than that of FQCs and SFs. However, excessive dye in cells may nonspecifically interact with cellular components to produce high background fluorescence. Considering the varied membrane permeability of the fluorogenic dye for different types of cell, careful optimization of the concentrations of fluorogenic dye for labeling is needed to reduce the background fluorescence and improve the S/B ratios (Figure 3D). Notably, several fluorogenic dyes, including Hoechst and Thiazole Orange derivatives, are nucleic acid dyes that nonspecifically interact with cellular nucleic acids to generate high background fluorescence [62]; thus, they are not suitable for RNA imaging in live cells. In addition, a covalent FR in which the fluorogen is covalently linked to the aptamer was recently reported, allowing the excessive dyes to be washed away to reduce the background fluorescence [63].

Photostability

FRs with excellent photostability are particularly useful for advanced RNA imaging. As described above, several FRs, such as Pepper:HBC620, Okra:ACE, RhoBAST:TMR-DN, RhoBAST:SpyRho, and biRhoBAST, have superb photostability, allowing real-time imaging of RNA dynamics at the single molecule level or with super-resolution [26–28,34,35,39]. However, several molecular rotor dyes, such as DFHBI and HBC530, undergo fast light-induced *cis*–*trans* photoisomerization

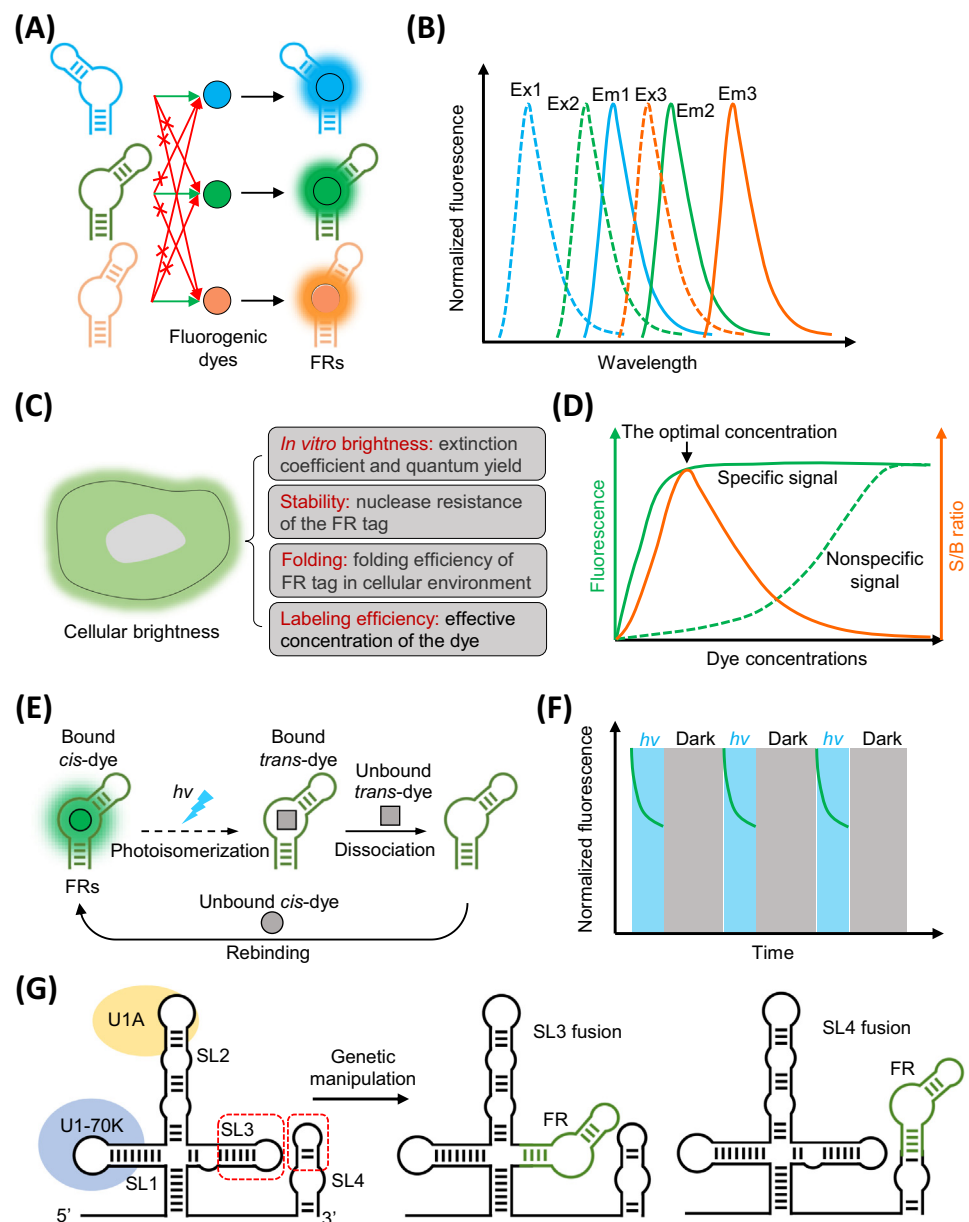


Figure 3. Considerations for enhancing the effectiveness of fluorescent RNAs (FRs). (A) No cross-interaction among the aptamers and fluorogenic dyes for orthogonal FRs. Red arrows represent no interaction between the aptamer and fluorogenic dye. (B) Minimal spectral overlap among orthogonal FRs to reduce false-positive signals caused by bleed-through. (C) Factors that influence cellular brightness. (D) Effects of fluorogenic dye concentrations on the specific and nonspecific fluorescent signals as well as the signal-to-background (S/B) ratios. (E) Schematic of the molecular steps that account for photobleaching of molecular rotor dye FRs and the restoration of fluorescence. (F) Unique photobleaching property of molecular rotor dye-based FRs. The fluorescence drops upon light illumination but recovers after removal of light. (G) U1 small nuclear (sn)RNAs as an example to show the strategy of genetic manipulation for any given RNA of interest (ROI). Red-dashed boxes indicate the sites tolerant of fusion with FR tags.

to a low-fluorescence *trans* form that subsequently dissociates from the aptamer, leading to loss of fluorescence. Unlike the permanent loss of fluorescence for FPs after photobleaching, the fluorescence of these FRs can be restored when a *cis*-dye from the solution rebinds the aptamer (Figure 3E,F) [64]. This unique property hinders the use of these FRs for continuous or super-resolution RNA imaging. However, the photobleaching of FRs can be suppressed by reducing the susceptibility of the dyes to photoisomerization or increasing the dissociation of the *trans*-dye and rebinding of the *cis*-dye to the aptamer; this can significantly increase the photostability of the FRs [23,28]. In addition to the FR itself, a pulsed illumination strategy was also developed to enhance the performance of these FRs for live cell RNA imaging [65].

Genetic manipulation

For visualization of an ROI using an FR tag, genetic manipulation, including fusion of the FR tag to the biologically compatible regions of the ROI, is needed. For mRNA imaging, FR tags are usually fused to the 3' UTR of mRNA transcripts, and their fusing has minimal effects on mRNA behavior, including localization, translation, and degradation [23,26,35]. By comparison, FR-based imaging of ncRNAs, particularly small ncRNAs, is not as simple as that of mRNAs, since snRNAs tend to form functional tertiary structures that interact with RBPs or RNAs; thus, most regions in snRNAs are not suitable for the fusion of exogenous FR tags [66]. In one example, human U1 snRNA forms four SLs (SL1–4). SL1 and SL2 directly interact with the U1-70K and U1-A RBPs, respectively, whereas SL3 and SL4 do not interact with RBPs and have only structural roles in maintaining the correct folding of U1 snRNA [67]. Thus, SL3 and SL4 can be replaced with FR tags for U1 snRNA labeling (Figure 3G) [27]. Therefore, the strategy of genetic manipulation for any given ROI needs to be carefully evaluated to minimize any possible artifacts. Notably, the FR-derived probes provide an alternative approach for RNA imaging if no suitable regions are available for fusion in the ROIs (Box 2).

Concluding remarks

With the emergence of FRs over the past two decades, direct tagging and imaging of RNA in live cells has become available to the RNA research community. As discussed here, FRs have been used for a range of applications, such as ncRNA and mRNA imaging, single molecule and super-resolution RNA imaging, and RNA–protein interactions in live mammalian cells. Furthermore, orthogonal FRs enable multiplexed fluorescence imaging to simultaneously visualize multiple RNA targets; this type of imaging is critical for understanding the correlations among different RNA species that are involved in cellular processes.

Since their first use for biological imaging in 1994 [68,69], numerous FPs have been developed, but several FPs with improved photophysical properties have been reported in recent years [70–73]. However, there is still much room for improvement in FRs in terms of their sensitivity, biocompatibility, and *in vivo* imaging capability. The abundance of cellular RNA is two to four orders of magnitude lower than that of protein [74]; thus, RNA visualization by FRs is not as simple as protein visualization by FP. Unlike most FPs, the chromophores of which are derived from internal amino acid residues, fluorogenic dyes are chemically synthesized and exogenously added into the cells. Owing to the background fluorescence of fluorogenic dyes, the sensitivity of FRs is typically lower than that of FPs. Therefore, the rational design of fluorogenic dyes with minimal background fluorescence and high cellular brightness upon aptamer binding is crucial for the development of novel FRs with enhanced imaging sensitivity. Typically, charged dyes have the tendency to nonspecifically interact with nucleic acids or acidic organelles (e.g., lysosome), and dyes with high hydrophobicity or polarity can interact with cellular membrane; these dyes normally have high background fluorescence. We recommend that the cellular background fluorescence of fluorogenic dyes is assessed before performing systematic evolution of ligands by exponential

Outstanding questions

Can FRs be used for real-time monitoring of the entire 'life cycle' of RNA molecules from transcription, splicing, modification, folding, and transportation to translation and degradation?

How can fluorogenic dyes with high fluorogenicity, and minimal nonspecific interactions and cellular components be rationally designed?

Can fluorogenic dyes be rationally designed to be bound by the tertiary structures of endogenous RNAs?

What is the minimum size of RNA aptamers that can still form tertiary structures capable of binding and activating the fluorescence of the fluorogenic dyes?

Artificial intelligence (AI) has been widely used for protein design. Can AI be applied to accelerate the development of novel customized FRs?

How FRs can be applied in live animals for the study of RNA dynamics? The answer to this question may provide new insights into RNA function.

enrichment (SELEX) screening. Moreover, since FR tags are biologically foreign to RNAs, they need to be as small as possible to reduce the possibility of interfering with the biological function of RNA. However, many currently available FRs are relatively large, with >50 nucleotides. Thus, smaller and more compact FRs with good biocompatibility with RNAs need to be developed. Furthermore, *in vivo* visualization of RNA dynamics can provide new insights into RNA function; however, this has not yet been attained with FRs. The main challenges include high autofluorescence, severe light scattering, low penetration of tissues, diffused distributions, and being easily metabolized. Recent progress in expanding the spectra of FRs into the near-infrared range can help improve the S/N ratios from RNA imaging *in vivo* [25]. In addition, the recently reported strategy for engineering covalent small molecule–RNA complexes would enable the development of a series of covalent FRs through attaching simple reactive handles to the fluorogenic dyes that enable covalent bonding with a guanosine nucleobase at the binding pocket [63], which may avoid a decrease in fluorescent signals as the dyes are metabolized (see [Outstanding questions](#)). Notably, covalent FRs will also be useful for the study of RNA turnover in live cells. To attain these FRs, close collaboration among biologists, chemists, physicists, and data analysts is crucial.

In addition to fluorogenic dyes, fluorogenic proteins whose fluorescence is activated by RNA aptamers have also been developed [5,75–77]. In these systems, fluorogenic proteins are highly unstable until they bind RNA motifs (e.g., RNA aptamers or SLs) inserted into target RNAs to generate FR–protein complexes. These systems exhibit significantly reduced background fluorescence and allow for live cell imaging of mRNA at the single molecule level with enhanced signal-to-background ratios. While appearing promising, these protein-based approaches are suboptimal for functional studies of single RNAs owing to the potential perturbation to RNA stability or behaviors caused by the bulky tag size and multiprotein tethering [35]. Moreover, real-time tracking of single endogenous RNA is useful for revealing both their spatial and temporal dynamics in live cells. FR- or fluorogenic protein-based approaches require genetic engineering of genomic DNA, which is always time-consuming and may interfere with native RNA behaviors. Alternatively, approaches using molecule beacons [78] or RNA-targeting CRISPR–Csm complexes [9] allow visualization of unmodified endogenous RNA at the single molecule level. However, they still suffer from the inability of the molecule beacons to be genetically encoded and the potential perturbation to RNA behaviors due to the tethering of multiple Csm complexes [9]. Thus, there is still a demand for the development of a live cell, single molecule, RNA imaging platform that is genetically encodable and protein binding independent, and that requires no genetic modification of target RNA.

Since robust imaging of RNA is needed to investigate its dynamics and biological function, scientists and creative minds will clearly have an important role in shaping the future of FRs. Currently, advances in FR technology have marked the beginning of a new chapter, not only for various fields in basic life science research in which RNAs are involved, but also for facilitating the development of advanced technologies for biosensing and diagnosis.

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Declaration of interests

The authors declare no competing interests.

References

- Chen, L.L. and Kim, V.N. (2024) Small and long non-coding RNAs: past, present, and future. *Cell* 187, 6451–6485
- Le, P. *et al.* (2022) Illuminating RNA biology through imaging. *Nat. Chem. Biol.* 24, 815–824
- Brasemann, E. *et al.* (2020) Illuminating RNA biology: tools for imaging RNA in live mammalian cells. *Cell Chem. Biol.* 27, 891–903
- Bertrand, E. *et al.* (1998) Localization of ASH1 mRNA particles in living yeast. *Mol. Cell* 2, 437–445
- Wu, J. *et al.* (2019) Live imaging of mRNA using RNA-stabilized fluorogenic proteins. *Nat. Methods* 16, 862–865
- Nelles, D.A. *et al.* (2016) Programmable RNA tracking in live cells with CRISPR/Cas9. *Cell* 165, 488–496
- Batra, R. *et al.* (2017) Elimination of toxic microsatellite repeat expansion RNA by RNA-targeting Cas9. *Cell* 170, 899–912
- Yang, L.Z. *et al.* (2019) Dynamic imaging of RNA in living cells by CRISPR-Cas13 systems. *Mol. Cell* 76, 981–997
- Xia, C. *et al.* (2025) Single-molecule live-cell RNA imaging with CRISPR-Csm. *Nat. Biotechnol.*, Published online February 18, 2025. <https://doi.org/10.1038/s41587-024-02540-5>
- Haimovich, G. *et al.* (2017) Inter-cellular mRNA trafficking via membrane nanotube-like extensions in mammalian cells. *Proc. Natl. Acad. Sci. U. S. A.* 114, E9873–E9882
- Garcia, J.F. and Parker, R. (2015) MS2 coat proteins bound to yeast mRNAs block 5' to 3' degradation and trap mRNA decay products: implications for the localization of mRNAs by MS2-MCP system. *RNA* 21, 1393–1395
- Paige, J.S. *et al.* (2011) RNA mimics of green fluorescent protein. *Science* 333, 642–646
- Lu, X. *et al.* (2023) Harmonizing the growing fluorogenic RNA aptamer toolbox for RNA detection and imaging. *Chem. Soc. Rev.* 52, 4071–4098
- Sando, S. *et al.* (2008) Transcription monitoring using fused RNA with a dye-binding light-up aptamer as a tag: a blue fluorescent RNA. *Chem. Commun. (Camb.)* 33, 3858–3860
- Dolgosheina, E.V. *et al.* (2014) RNA mango aptamer-fluorophore: a bright, high-affinity complex for RNA labeling and tracking. *ACS Chem. Biol.* 9, 2412–2420
- Tan, X. *et al.* (2017) Fluoromolecules consisting of a promiscuous RNA aptamer and red or blue fluorogenic cyanine dyes: selection, characterization, and bioimaging. *J. Am. Chem. Soc.* 139, 9001–9009
- Cawte, A.D. *et al.* (2020) Live cell imaging of single RNA molecules with fluorogenic Mango II arrays. *Nat. Commun.* 11, 1283
- Strack, R.L. *et al.* (2013) A superfolder Spinach2 reveals the dynamic nature of trinucleotide repeat-containing RNA. *Nat. Methods* 10, 1219–1224
- Song, W. *et al.* (2014) Plug-and-play fluorophores extend the spectral properties of Spinach. *J. Am. Chem. Soc.* 136, 1198–1201
- Filonov, G.S. *et al.* (2014) Broccoli: rapid selection of an RNA mimic of green fluorescent protein by fluorescence-based selection and directed evolution. *J. Am. Chem. Soc.* 136, 16299–16308
- Song, W. *et al.* (2017) Imaging RNA polymerase III transcription using a photostable RNA-fluorophore complex. *Nat. Chem. Biol.* 13, 1187–1194
- Steinmetzger, C. *et al.* (2019) A multicolor large Stokes shift fluorogen-activating RNA aptamer with cationic chromophores. *Chemistry* 25, 1931–1935
- Li, X. *et al.* (2020) Fluorophore-promoted RNA folding and photostability enables imaging of single Broccoli-tagged mRNAs in live mammalian cells. *Angew. Chem. Int. Ed. Engl.* 59, 4511–4518
- Li, X. *et al.* (2021) Engineering fluorophore recycling in a fluorogenic RNA aptamer. *Angew. Chem. Int. Ed. Engl.* 60, 24153–24161
- Chen, Z.Y. *et al.* (2025) Near-infrared fluorogenic RNA for in vivo imaging and sensing. *Nat. Commun.* 16, 518
- Chen, X. *et al.* (2019) Visualizing RNA dynamics in live cells with bright and stable fluorescent RNAs. *Nat. Biotechnol.* 37, 1287–1293
- Jiang, L. *et al.* (2023) Large Stokes shift fluorescent RNAs for dual-emission fluorescence and bioluminescence imaging in live cells. *Nat. Methods* 20, 1563–1572
- Zuo, F. *et al.* (2024) Imaging the dynamics of messenger RNA with a bright and stable green fluorescent RNA. *Nat. Chem. Biol.* 20, 1272–1281
- Jiang, L. *et al.* (2024) Bright and stable cyan fluorescent RNA enables multicolor RNA imaging in live *Escherichia coli*. *Small* 21, e2405165
- Sunbul, M. and Jaschke, A. (2013) Contact-mediated quenching for RNA imaging in bacteria with a fluorophore-binding aptamer. *Angew. Chem. Int. Ed. Engl.* 52, 13401–13404
- Arora, A. *et al.* (2015) Dual-colour imaging of RNAs using quencher- and fluorophore-binding aptamers. *Nucleic Acids Res.* 43, e144
- Sunbul, M. and Jaschke, A. (2018) SRB-2: a promiscuous rainbow aptamer for live-cell RNA imaging. *Nucleic Acids Res.* 46, e110
- Brasemann, E. *et al.* (2018) A multicolor riboswitch-based platform for imaging of RNA in live mammalian cells. *Nat. Chem. Biol.* 14, 964–971
- Sunbul, M. *et al.* (2021) Super-resolution RNA imaging using a rhodamine-binding aptamer with fast exchange kinetics. *Nat. Biotechnol.* 39, 686–690
- Buhler, B. *et al.* (2023) Avidity-based bright and photostable light-up aptamers for single-molecule mRNA imaging. *Nat. Chem. Biol.* 19, 478–487
- Bouhedda, F. *et al.* (2020) A dimerization-based fluorogenic dye-aptamer module for RNA imaging in live cells. *Nat. Chem. Biol.* 16, 69–76
- Wirth, R. *et al.* (2019) SIRA: a silicon rhodamine-binding aptamer for live-cell super-resolution RNA imaging. *J. Am. Chem. Soc.* 141, 7562–7571
- Zhang, J. *et al.* (2021) A color-shifting near-infrared fluorescent aptamer-fluorophore module for live-cell RNA imaging. *Angew. Chem. Int. Ed. Engl.* 60, 21441–21448
- Englert, D. *et al.* (2023) Fast-exchanging spirocyclic rhodamine probes for aptamer-based super-resolution RNA imaging. *Nat. Commun.* 14, 3879
- Zheng, R. *et al.* (2024) Multiplexed sequential imaging in living cells with orthogonal fluorogenic RNA aptamer/dye pairs. *Nucleic Acids Res.* 52, e67
- Wang, Q. *et al.* (2022) Inert Pepper aptamer-mediated endogenous mRNA recognition and imaging in living cells. *Nucleic Acids Res.* 50, e84
- Zheng, H. *et al.* (2025) Imaging of endogenous RNA in live cells using sequence-activated fluorescent RNA probes. *Nucleic Acids Res.* 53, gkae1209
- Tang, A.A. *et al.* (2024) Optimization of RNA Pepper sensors for the detection of arbitrary RNA targets. *ACS Synth. Biol.* 13, 498–508
- Ong, W.Q. *et al.* (2017) Live cell imaging of endogenous mRNA using RNA-based fluorescence 'Turn-On' probe. *ACS Chem. Biol.* 12, 200–205
- Yin, P. *et al.* (2023) A universal orthogonal imaging platform for living-cell RNA detection using fluorogenic RNA aptamers. *Chem. Sci.* 14, 14131–14139
- Huang, Z. *et al.* (2023) Genetically encodable tagging and sensing systems for fluorescent RNA imaging. *Biosens. Bioelectron.* 219, 114769
- Truong, L. and Ferre-D'Amare, A.R. (2019) From fluorescent proteins to fluorogenic RNAs: tools for imaging cellular macromolecules. *Protein Sci.* 28, 1374–1386
- Mysara, M. *et al.* (2011) MysiRNA-designer: a workflow for efficient siRNA design. *PLoS One* 6, e25642
- Autour, A. *et al.* (2018) Fluorogenic RNA Mango aptamers for imaging small non-coding RNAs in mammalian cells. *Nat. Commun.* 9, 656
- Dey, S.K. *et al.* (2022) Repurposing an adenine riboswitch into a fluorogenic imaging and sensing tag. *Nat. Chem. Biol.* 18, 180–190
- Tsalikis, J. *et al.* (2015) Intracellular bacterial pathogens trigger the formation of U small nuclear RNA bodies (U Bodies) through metabolic stress induction. *J. Biol. Chem.* 290, 20904–20918
- Jain, S. *et al.* (2016) ATPase-modulated stress granules contain a diverse proteome and substructure. *Cell* 164, 487–498

53. Huang, B. *et al.* (2010) Breaking the diffraction barrier: super-resolution imaging of cells. *Cell* 143, 1047–1058
54. Yin, P. *et al.* (2025) Developing orthogonal fluorescent RNAs for photoactive dual-color imaging of RNAs in live cells. *Angew. Chem. Int. Ed. Engl.* 64, e202424060
55. Lippincott-Schwartz, J. and Patterson, G.H. (2009) Photoactivatable fluorescent proteins for diffraction-limited and super-resolution imaging. *Trends Cell Biol.* 19, 555–565
56. Ramanathan, M. *et al.* (2019) Methods to study RNA-protein interactions. *Nat. Methods* 16, 225–234
57. Ramanathan, M. *et al.* (2018) RNA-protein interaction detection in living cells. *Nat. Methods* 15, 207–212
58. Graindorge, A. *et al.* (2019) In-cell identification and measurement of RNA-protein interactions. *Nat. Commun.* 10, 5317
59. Lorenz, M. (2009) Visualizing protein-RNA interactions inside cells by fluorescence resonance energy transfer. *RNA* 15, 97–103
60. Huranova, M. *et al.* (2009) In vivo detection of RNA-binding protein interactions with cognate RNA sequences by fluorescence resonance energy transfer. *RNA* 15, 2063–2071
61. Mohanty, S.K. and Lakshminarayanan, V. (2015) Optical techniques in optogenetics. *J. Mod. Opt.* 62, 949–970
62. Suss, O. *et al.* (2021) Broad applications of thiazole orange in fluorescent sensing of biomolecules and ions. *Molecules* 26, 2828
63. Bereiter, R. *et al.* (2025) Engineering covalent small molecule-RNA complexes in living cells. *Nat. Chem. Biol.*, Published online January 6, 2025. <https://doi.org/10.1038/s41589-024-01801-3>
64. Han, K.Y. *et al.* (2013) Understanding the photophysics of the spinach-DFHBI RNA aptamer-fluorogen complex to improve live-cell RNA imaging. *J. Am. Chem. Soc.* 135, 19033–19038
65. Zhang, J. *et al.* (2015) Tandem Spinach array for mRNA imaging in living bacterial cells. *Sci. Rep.* 5, 17295
66. Tholen, J. and Galej, W.P. (2022) Structural studies of the spliceosome: bridging the gaps. *Curr. Opin. Struct. Biol.* 77, 102461
67. Pomeranz Krummel, D.A. *et al.* (2009) Crystal structure of human spliceosomal U1 snRNP at 5.5 Å resolution. *Nature* 458, 475–480
68. Chalfie, M. *et al.* (1994) Green fluorescent protein as a marker for gene expression. *Science* 263, 802–805
69. Service, R.F. (2008) Nobel Prize in chemistry. Three scientists bask in prize's fluorescent glow. *Science* 322, 361
70. Zhang, H. *et al.* (2024) Bright and stable monomeric green fluorescent protein derived from StayGold. *Nat. Methods* 21, 657–665
71. Ivorra-Molla, E. *et al.* (2024) A monomeric StayGold fluorescent protein. *Nat. Biotechnol.* 42, 1368–1371
72. Gadella, T.W.J., Jr. *et al.* (2023) mScarlet3: a brilliant and fast-maturing red fluorescent protein. *Nat. Methods* 20, 541–545
73. Xiong, H. *et al.* (2025) A highly stable monomeric red fluorescent protein for advanced microscopy. *Nat. Methods*, Published online April 17, 2025. <https://doi.org/10.1038/s41592-025-02676-5>
74. Schwanhauser, B. *et al.* (2011) Global quantification of mammalian gene expression control. *Nature* 473, 337–342
75. Zhou, W.J. *et al.* (2025) Fluorogenic interacting protein stabilization for orthogonal RNA imaging. *Angew. Chem. Int. Ed. Engl.*, Published online April 10, 2025. <https://doi.org/10.1002/anie.202502350>
76. Kuffner, C.J. *et al.* (2024) RNA-stabilized coat proteins for sensitive and simultaneous imaging of distinct single mRNAs in live cells. *bioRxiv*, Published online November 21, 2024. <https://doi.org/10.1101/2024.11.21.624393>
77. Pham, T.G. *et al.* (2024) Orthogonal RNA-regulated destabilization domains for three-color RNA imaging with minimal RNA perturbation. *bioRxiv*, Published online December 20, 2024. <https://doi.org/10.1101/2024.12.16.628815>
78. Mao, S. *et al.* (2020) Recent advances in the molecular beacon technology for live-cell single-molecule imaging. *iScience* 23, 101801
79. Chu, J. *et al.* (2016) A bright cyan-excitable orange fluorescent protein facilitates dual-emission microscopy and enhances bioluminescence imaging in vivo. *Nat. Biotechnol.* 34, 760–767
80. Platkevich, K.D. *et al.* (2010) Monomeric red fluorescent proteins with a large Stokes shift. *Proc. Natl. Acad. Sci. U. S. A.* 107, 5369–5374
81. Yang, J. *et al.* (2013) mBeRFP, an improved large Stokes shift red fluorescent protein. *PLoS One* 8, e64849
82. Wu, J. *et al.* (2014) A long Stokes shift red fluorescent Ca²⁺ indicator protein for two-photon and ratiometric imaging. *Nat. Commun.* 5, 5262
83. Ying, Z.M. *et al.* (2017) Genetically encoded fluorescent RNA sensor for ratiometric imaging of microRNA in living tumor cells. *J. Am. Chem. Soc.* 139, 9779–9782
84. Yuan, D. *et al.* (2025) Allosteric genetically encoded biosensor for spatiotemporal monitoring of endogenous RNA dynamics in living cells. *Proc. Natl. Acad. Sci. U. S. A.* 122, e2409309122
85. Buhler, B. *et al.* (2023) Programmable, structure-switching RhoBAST for hybridization-mediated mRNA imaging in living cells. *ACS Chem. Biol.* 18, 1838–1845
86. Pothoulakis, G. *et al.* (2014) The spinach RNA aptamer as a characterization tool for synthetic biology. *ACS Synth. Biol.* 3, 182–187
87. Sarfraz, N. *et al.* (2023) Visualizing orthogonal RNAs simultaneously in live mammalian cells by fluorescence lifetime imaging microscopy (FLIM). *Nat. Commun.* 14, 867